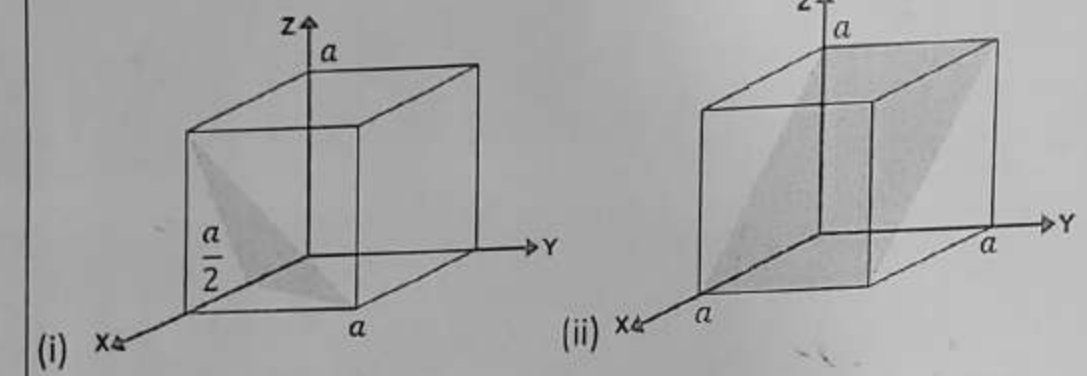
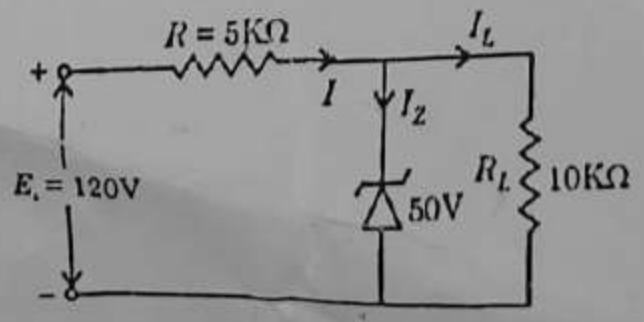
(Please write your Enrollment Number) Enrollment No. 02410100023

MID-TERM EXAMINATION
(Course Name:B. Tech. CSE/ECE/IT/MAE/DMAM) (Semester: Second)
(March, 2024) OFF LINE mode

Subject Code: BAS 104	**Subject: Applied Physics II**
Time:1 ½Hours	**Maximum Marks:30**
Note:Q. 1 is compulsory.	

Q1		**(2.5*4)**	
	(a) A parallel-plate capacitor with circular plates of radius $a = 0.055$m is being charged at a uniform rate so that the electric field between the plates changes at a constant rate $\frac{\partial E}{\partial t} = 1.5 \times 10^{13}$ V/m/s Determine the displacement current for the capacitor.	UNIT-1	CO 1
	(b) The potential of a certain electrostatic field has the form $V = -a(x^2 + y^2) + bz^2$, where a and b are constants. Find the magnitude and direction of the electric field vector.	UNIT-1	
	(c) Assuming a simple cubic lattice of side equal toa, determine the Miller Indices of the given lattice planes: 	UNIT-2	CO 2
	(d)State and explain Bloch theorem for one dimensional periodic potential.	UNIT-2	
Q2	**(Attempt any Two Parts) UNIT-1 (CO 1)**	**(5,5)**	
	(a) Discuss in detail the electromagnetic wave propagation in a conducting medium. Why do we get very low mobile signal in a lift?		
	(b) Show that Ampere's law is inconsistent in time varying fields.Explain how Maxwell corrected and presented the Ampere's law in its generalized form.		
	(c) In a loss less medium, the relative permittivity and permeability have values equal to $\varepsilon_r = 4\mu_r = 1$. If the magnetic field is given as $\vec{H}(z,t) = -0.1\cos(\omega t - z)\hat{\imath} + 0.5\sin(\omega t - z)\hat{\jmath}$ A/m, find the value of angular frequency ω and the electric field $\vec{E}(z,t)$.		
Q3	**(Attempt any Two Parts) UNIT-2 (CO 2)**	**(5,5)**	
	(a) Differentiate between the following: (i) Intrinsic and Extrinsic Semiconductor (ii) Direct Lattice and Reciprocal lattice		
	(b) Find the value of currents $I, I_L, and\ I_z$ in the following circuit. 		
	(c) Discuss the Kronig-Penney model for the motion of electrons in a periodic potential. Explain the formation of allowed and forbidden energy bands using the $E - K$ diagram.		